Santiago, Chile
Available for on-site and/or remote work
+56 9 54039592
rodmolina.avila@gmail.com
rodrigomolinaavila.github.io/RodrigoMolinaAvila
rodmolinaavila
RodrigoMolinaAvila
ORCID iD: 0009-0006-0648-0799

Rodrigo Molina Ávila

Data Scientist | Computational Sociologist

Professional Profile

Data Scientist and Sociologist, MSc in Data Science from the Faculty of Physical and Mathematical Sciences at the Universidad de Chile, with 5+ years of experience developing analytical and predictive models applied to institutional, academic, and social phenomena. My work lies at the intersection of sociological training —deep understanding of gender, social class, elites, politics, and crisis— and the technical capacity to model them rigorously through knowledge graphs, NLP, machine learning, and data engineering. Specialized in transforming large volumes of heterogeneous data (relational, textual, and network) into actionable knowledge for strategic decision-making. Proven experience designing end-to-end pipelines: from data integration and quality control to executive dashboards and reports for technical and non-technical audiences. Methodological rigor, interdisciplinary communication, and reproducibility as working standards.

Professional Experience

March 2025 – September 2026 **Data Scientist — Thesis Researcher, Applied Research**, *Data and Artificial Intelligence Initiative (IDIA) — Millennium Institute for Foundational Research on Data (IMFD)*, Thesis advisor: Sebastián Ferrada, Universidad de Chile, Santiago, Chile · Institutional profile · Thesis presentation

- **Knowledge graphs:** designed and built a Neo4j knowledge graph with **16 node types** and **15 relationship types** —over **3,000,000 co-authorship edges**— modeling the academic trajectories and scientific output of **3,143 ANID doctoral fellows** (2014–2016 cohorts).
- **Data modeling:** federated seven heterogeneous sources (ANID, Scopus, Mineduc, QS and Shanghai rankings, CNA) with Python and SQL, performing entity resolution for author and institution disambiguation and normalization of administrative records.
- **Network analysis methodologies:** community detection with Louvain and Leiden, connected component analysis, and centrality metrics (PageRank, betweenness, closeness) on the intra-cohort co-authorship network, validated against null models (Erdős–Rényi, SBM).
- **Topic modeling: 237 topics** with BERTopic over 26,071 publications (multilingual SBERT, UMAP, HDBSCAN, and c-TF-IDF), validated with topic coherence, silhouette, and AMI.
- **Predictive and clustering models:** XGBoost, LightGBM, and CatBoost with Optuna, stratified cross-validation, and SHAP interpretability to explain academic success factors.
- **Visualization and outreach:** executive dashboards in Power BI and an interactive D3.js visualizer over Graph Data Science; public platform at interrogandolaacademia.cl.
- **Technologies:** Python, Neo4j (GDS, Cypher), SQL, scikit-learn, pandas, BERTopic, SBERT, XGBoost, LightGBM, CatBoost, SHAP, Power BI, D3.js, Docker, Git

March 2026 – May 2026 **Consultant — Qualitative Evaluation**, *Data and Artificial Intelligence Initiative (IDIA)*, Professional Master's Program in Artificial Intelligence, FCFM, Santiago, Chile

- **Conducted a qualitative evaluation** of the relevance, curriculum design, and implementation conditions of the Professional Master's in AI, through a focus group with alumni and interviews with key industry stakeholders.
- **Built a complete analysis pipeline**: speaker diarization, transcription, thematic coding of the corpus, and natural language processing over the structured corpus.
- **Developed a navigable web interface** exposing the full chain of evidence by objective and participant, guaranteeing traceability of the study.
- **Technologies:** Python, audio diarization, NLP, LLMs, HTML/JavaScript

May 2025 **Quantitative Consultant**, *Fab Consulting*, Evaluation of the Research Valorization Program, ANID, Santiago, Chile

- Technical proposal prepared for the Undersecretariat of Science, Technology, Knowledge and Innovation.
- Quantitative analysis of the program's results.

October 2024 – May 2025 **Research Assistant**, *Center for Social Conflict and Cohesion Studies (COES)*, Chilean Think Tanks in Times of Crisis, Santiago, Chile

- **Developed a data extraction and analysis system** for **20+ institutions**, processing **40,000+ documents** with automated web scraping and NLP pipelines, published on **GitHub Pages**.
- **Modeled influence networks** using topic analysis (BERTopic) and network analysis, identifying production patterns and temporal trends.
- **Trained the research team** in R, Python, and Git for reproducible data analysis.
- **In progress**: academic paper in computational sociology journals.
- **Technologies:** Python, R, SQL, Web Scraping, pandas, scikit-learn, GitHub Pages, D3.js, Power BI

September 2024 – May 2025 **Research Assistant**, *FONDECYT 11241304*, Inequalities in STEM Academia, Santiago, Chile

- **Designed a mixed-methods approach** to analyze academic trajectories with a gender and social class intersectional perspective on a population of **3,000+ researchers**.
- **Implemented clustering models** (K-Means, hierarchical clustering) and multiple correspondence analysis (MCA) for academic profile segmentation.
- **Produced interactive visualizations** to communicate complex findings to interdisciplinary teams and non-technical stakeholders.
- **Technologies:** Python, R, pandas, scikit-learn, applied statistics, Power BI

February 2023 – March 2024 **Research Assistant**, *FONDECYT POSTDOC 3210579*, Think Tanks amid Presidential Transitions, Santiago, Chile

- **Built a unified database** with **48,000+ publications** through automated extraction from APIs and official websites, integrating multi-source data.
- **Designed a relational model** and transformation pipelines ensuring data consistency and availability for longitudinal analysis.
- **Developed Power BI dashboards** for time-series production density analysis.
- **Technologies:** Python, R, SQL, Power BI, REST APIs

February 2022 – 2023 **Research Assistant**, *Mini COES — Constitutional Convention*, Expert networks: migration and decentralization, Santiago, Chile

- **Built databases** and conducted data mining on expert networks in the migration and decentralization debate.
- **Ethnography** and network generation inside the former National Congress.
- **Semi-structured interviews**, transcription, and coding in Atlas.ti.
- **Writing** and production of publications.
- **Technologies:** Atlas.ti, R, Excel

September 2021 – February 2022 **Technical Staff**, *FONDECYT Regular 11200602*, Geographic Inequality in a Double Register, Santiago, Chile

- **Systematization of categories** for qualitative discourse analysis in the Antofagasta and La Araucanía markets.
- **Grounded Theory**: theoretical analysis, transcription, and categorization.

2019 **Survey Taker**, *Datavoz*, Barómetro de la Política, Santiago, Chile

- Data collection for the Barómetro de la Política.

May 2018 – September 2018 **Professional Internship**, *Gendarmería de Chile*, Classification and Segmentation Department, National Directorate, Santiago, Chile

- **Methodological update** of the classification form: database analysis, statistical tests, and instrument design.
- **Fieldwork** inside Santiago 1 and CCP Puente Alto prisons: qualitative sampling and bibliographic analysis.

Education

March 2024 – 2026 **MSc in Data Science**, *Universidad de Chile*, Faculty of Physical and Mathematical Sciences (FCFM), Santiago, Chile

- **Specialization:** Machine Learning, Business Intelligence, Knowledge Graphs, Natural Language Processing, Causal Inference, Relational and Non-Relational Databases, Artificial Intelligence
- **Thesis:** Neo4j Knowledge Graph modeling the academic trajectories and scientific output of **3,143 ANID doctoral fellows** (2014–2016 cohorts) with BERTopic and Graph Data Science. Public platform: interrogandolaacademia.cl

2014 – 2019 **BA in Sociology (Sociologist)**, *Universidad Andrés Bello*, Faculty of Education and Social Sciences, Santiago, Chile

- **Specialization:** Quantitative and Qualitative Research, Applied Statistics, Public Policy Evaluation, Organizational Development

Technical Skills

Languages	Python, R, SQL (PostgreSQL, SQLite), Cypher (Neo4j), SPARQL
ML & Statistics	scikit-learn, XGBoost, LightGBM, CatBoost, SHAP, Optuna, statsmodels, Clustering, Dimensionality Reduction, Predictive Modeling, Cross-Validation, Community Detection (Louvain, Leiden), Centrality Measures (PageRank, Betweenness)
Data Engineering	pandas, NumPy, ETL Pipelines, Web Scraping, REST APIs, Multi-source Integration, Data Quality Control
Databases	SQL (PostgreSQL, SQLite), NoSQL, Neo4j, Property Graphs, RDF/OWL, Wikidata, Excel Ecosystem
NLP	Transformers, BERTopic, SBERT, Embeddings, Sentiment Analysis, spaCy, NLTK, Speaker Diarization, LLM-Assisted Qualitative Analysis
Viz & BI	Power BI, D3.js, Plotly, Matplotlib, Seaborn, GitHub Pages, HTML/CSS/JavaScript
MLOps & Deploy	MLflow, Docker, Git, Apache Airflow, Jupyter, Gradio
Harness Engineering & AI APIs	Claude Code, omp, MCP, OpenRouter, OpenDesign, DeepSeek harness, AI APIs, RAG, prompt engineering, local LLMs (Ollama), fine-tuning / LoRA

Publications

2026 **Untangling the Academic Network: Analyzing Scientific Trajectories with Graphs and Natural Language Processing**, *AMW 2026 (CEUR-WS)*, Accepted

- **First author:** Rodrigo Molina. **Co-authors:** Sebastián Ferrada and Roxana Chiappa.

2026 **Think Tanks and Political Crises in Chile, 2011–2022; or, The Crystallization of the Intellectual Infrastructure in a Middle-Income Country**, *Latin American Research Review (Cambridge University Press)*, M. González Hernando

- **Researcher acknowledged** for contributions to the Chilean think tank research line (FONDECYT POSTDOC 3210579 / COES). doi:10.1017/lar.2026.10142

Supervised Theses

August 2025 – October 2026 **Co-supervisor — Undergraduate Thesis**, *Web platform for access to data on science funding in Chile*, Student: Matías Vega — Department of Computer Science (DCC), Faculty of Physical and Mathematical Sciences, Universidad de Chile

- Co-supervision of the thesis: data architecture design and supervision of the public platform's development.

Teaching

August 2026 – December 2026 **Assistant Professor — Natural Language Processing (Andrés Abeliuk and Fabrizzio Pezzolla)**, *Diploma in Computer Science*, Universidad de Chile, Santiago, Chile

May 2026 – August 2026 **Teaching Assistant — Information Retrieval (IR) (Sebastián Ferrada)**, *FCFM Diploma Program*, Universidad de Chile, Santiago, Chile

March 2026 – July 2026 **Teaching Assistant — Knowledge Graphs (Sebastián Ferrada)**, *MSc in Data Science, FCFM*, Universidad de Chile, Santiago, Chile

March 2026 – July 2026 **Teaching Assistant — Fundamentals of Sociology (Marcos González Hernando)**, *Universidad Diego Portales*, Faculty of Sociology, Santiago, Chile

Second Semester 2022 **Teaching Assistant — Latin American Politics (Luis Garrido Vergara)**, *Faculty of Government, Universidad de Chile*, Political Science, Santiago, Chile

Second Semester 2022 **Teaching Assistant — Human Behavior in Organizations (Luis Garrido Vergara)**, *Faculty of Government, Universidad de Chile*, Political Science, Santiago, Chile

Second Semester 2022 **Teaching Assistant — Empirical Analysis of Public Policies (Juan Carlos Cortázar Velarde)**, *Faculty of Government, Universidad de Chile*, Public Administration, Santiago, Chile

Conferences

August 2026 **IV FCFM Graduate Symposium**, *Trajectories of ANID Fellows through Knowledge Graphs and Natural Language Processing: what public investment produces and how its scientific community is structured*, Universidad de Chile, Santiago, Chile
- **Co-author with Sebastián Ferrada (IDIA – IMFD).** **View poster**

January 2026 **VIII IMFD Workshop**, *Knowledge graph representation of academic trajectories and scientific output of Becas Chile and CONICYT doctoral fellows across disciplines, topics, and co-authorships differentiated by gender and social class*, Millennium Institute for Foundational Research on Data (IMFD), Santiago, Chile
- **Speaker.** **View poster**

December 2025 **Summer School on Computational Intelligence (EVIC 2025)**, *Participant*, Universidad de los Andes, Santiago, Chile
- Participation in the summer school on computational intelligence.

November 2025 **Research Findings Presentation**, *Chilean Think Tanks in Turbulent Times: 2019–2023*, Universidad Alberto Hurtado, Santiago, Chile
- **Speaker:** MiniCOES research findings with an interdisciplinary team. **View presentation**

October 2025 **XII COES International Conference**, *Is Democracy Really the Problem?*, COES, Santiago, Chile
- **Presenter:** "Expert knowledge in turbulent times: think tanks during Chile's political crises (2019–2023)"
- **View presentation**

November 2022 **IX COES International Conference**, *The Future of the Commons*, COES, Santiago, Chile
- **Presenter:** "The Role of Expert Knowledge in the Constitutional Convention"

Languages

Spanish	Native
English	Advanced (academic reading, writing, and conversation)
Chinese (中文)	Basic